TSGD: The Sister of SGD

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Abstract

Stochastic gradient descent (SGD) estimates gradients. Transported SGD (TSGD) estimates their changes and uses them to update an initial gradient estimate along the optimization trajectory. Estimating changes can be much easier than estimating gradients: even when individual gradients are highly noisy, shared additive noise cancels exactly from their differences, giving exact transport. With exact initialization as well, TSGD follows gradient descent exactly, regardless of the noise magnitude. This known recursion appears in several variance-reduced methods; here we study it directly as a fundamental estimation mechanism. Its accuracy depends on two error sources: initialization error, which persists, and transport errors, which accumulate. Centered similarity controls the latter through noise variation rather than pointwise noise magnitude. We give a clean, unified convergence analysis that separates these contributions, and derive exact quadratic dynamics that quantify how initialization and transport errors interact. Simple examples and least-squares experiments identify regimes favoring TSGD or SGD. These results clarify both the value of pure transport and why modern optimizers combine it with fresh estimation.

1 Introduction

Stochastic gradient descent (SGD) estimates gradients afresh. Transported SGD (TSGD) estimates gradient changes and uses them to update an existing estimate. SGD is a cornerstone of stochastic optimization and a workhorse of machine learning: replacing the exact gradient with a stochastic estimate makes each update simple and scalable [1]. TSGD follows an equally natural principle: it adds estimated gradient changes to its current estimate, transporting information along the optimization trajectory.

Estimating changes can be much easier than estimating absolute quantities. Subtracting gradients evaluated with the same sample cancels their shared noise component. In particular, shared additive noise cancels exactly, giving the true gradient change even when individual gradients are highly noisy. With exact initialization as well, TSGD follows gradient descent path by path, regardless of the noise magnitude.

The two methods therefore depend on different aspects of the noise. Classical SGD guarantees depend on its pointwise magnitude, commonly bounded by a variance parameter σ^2 [6]. TSGD guarantees, on the other hand, depend on how the noise varies between the coupled evaluations along the trajectory. Centered δ-similarity quantifies this variation: small δ means that stochastic gradient changes closely match deterministic gradient changes. Thus, transport can remain accurate even when pointwise noise is large. We stress that centered similarity applies to population objectives as well as finite sums.

Two sources of error determine TSGD's accuracy: error in the initial gradient estimate and error in the estimated gradient changes. The initial error persists; new transport errors accumulate along the trajectory. Accurate differences preserve a good initial estimate, but cannot correct an inaccurate one. The TSGD guarantee therefore depends on both δ and the accuracy of the initial

estimate. At fixed values of these quantities, it is independent of σ, although the noise magnitude can affect the cost of obtaining an accurate initial estimate.

Least-squares regression provides a concrete example. Label noise enters the stochastic gradient $u(u^\top x - b)$, but cancels from the shared-sample difference $uu^\top(x-y)$. It therefore affects fresh gradient estimates and TSGD's initialization, but not its subsequent transport increments (Section 2.2).

The transport recursion is not new. It appears in established variance-reduced algorithms such as SARAH, SPIDER, and PAGE [4, 12, 13], and independent single-run analyses already exist [14, 15, 18]. It is commonly encountered as part of methods that combine transport with refreshing or mixing. Our purpose is to study the mechanism directly, making its strengths and limitations as explicit as those of SGD.

Momentum SGD (MSGD) provides a third fundamental mechanism: averaging fresh gradients with past estimates. Section 2 introduces all three; Section 5 discusses their combination in modern optimizers.

The paper develops this viewpoint through three contributions:

- **Fundamental viewpoint.** We present TSGD alongside SGD as a basic way of using stochastic-gradient information: estimating changes rather than absolute gradients. Similarity explains when these changes can remain accurate despite noisy individual gradients.
- **Theory and understanding.** We explain similarity, separate initialization and transport errors, and give a finite-horizon guarantee under conditional assumptions on the gradient change increments. Exact quadratic dynamics quantify their interaction and show how the estimator can vanish while the true gradient does not.
- **Comparison and connections.** We construct oracles with identical pointwise gradient distributions but different transport accuracy, and give exact examples showing when SGD or TSGD has the smaller stationarity error. Least-squares experiments illustrate these regimes and the effect of initialization. We compare convergence guarantees and oracle costs, and identify how established optimizers combine transport with fresh estimation and momentum averaging.

The examples precede the convergence analysis so that the role of each assumption is visible before the theorem. Detailed proofs and additional calculations appear in the appendices.

2 Three Fundamental Estimators and Similarity

2.1 SGD, MSGD, and TSGD

After taking a step to x_{k+1}, we need a gradient estimate to determine the next update. The identity

$$\nabla f(x_{k+1}) = \nabla f(x_k) + [\nabla f(x_{k+1}) - \nabla f(x_k)]$$

suggests two approaches. SGD estimates the left-hand side directly by drawing a fresh stochastic gradient at the new point. TSGD estimates the right-hand side: its current estimate g_k already approximates the first term, so it needs new information only for the change. Using a stochastic gradient oracle h with the same fresh sample at both endpoints gives

$$g_{k+1} = \underbrace{g_k}_{\approx \nabla f(x_k)} + \underbrace{h(x_{k+1}, \xi_{k+1}) - h(x_k, \xi_{k+1})}_{\approx \nabla f(x_{k+1}) - \nabla f(x_k)}.$$

Thus TSGD updates the gradient information it already has. The sample ξ_{k+1} is shared within the difference and drawn afresh at each iteration. For a finite sum, this means evaluating the same randomly selected component at both points. TSGD retains the information in g_k, including its estimation error; the effect of these inherited errors is examined in Section 2.3.

We formalize the setup as follows.

Assumption A1 (Smooth, lower-bounded objective)**.** The function $f : \mathbb{R}^d \to \mathbb{R}$ is lower bounded, $f_{\inf} := \inf_x f(x) > -\infty$, and has an L-Lipschitz gradient with $L > 0$:

$$\|\nabla f(x) - \nabla f(y)\| \le L\|x - y\| \qquad \text{for all } x, y \in \mathbb{R}^d.$$

Assumption A2 (Unbiased stochastic gradients)**.** The oracle satisfies $\mathbb{E}_\xi[h(x, \xi)] = \nabla f(x)$ for every x. Each new sample is independent of the preceding history.

All three methods use positive deterministic stepsizes η_k and the same iterate update, starting from a deterministic x_0:

$$x_{k+1} = x_k - \eta_k g_k. \tag{2.1}$$

Their gradient estimates differ:

$$\text{SGD}: \quad g_{k+1} = h(x_{k+1}, \xi_{k+1}), \tag{2.2}$$

$$\text{MSGD}: \quad g_{k+1} = (1 - \beta_k)g_k + \beta_k h(x_{k+1}, \xi_{k+1}), \quad 0 < \beta_k \le 1, \tag{2.3}$$

$$\text{TSGD}: \quad g_{k+1} = g_k + D_{k+1}. \tag{2.4}$$

In the standard pointwise-oracle construction,

$$D_{k+1} = h(x_{k+1}, \xi_{k+1}) - h(x_k, \xi_{k+1}). \tag{2.5}$$

SGD replaces its estimate with a fresh stochastic gradient at every iteration. MSGD averages fresh and past estimates to reduce noise, but can lag behind a changing gradient. TSGD accounts for the gradient change through transport. Unless specified otherwise, SGD and MSGD initialize with $g_0 = h(x_0, \xi_0)$. TSGD permits any g_0 with finite mean-square error, including a biased estimate.

2.2 Understanding Similarity

SGD is typically analyzed under a bounded-variance assumption. Write $r(x, \xi) = h(x, \xi) - \nabla f(x)$ for the gradient noise.

Assumption A3 (Bounded variance (BV))**.** For some $\sigma \ge 0$,

$$\mathbb{E}_\xi[\|r(x, \xi)\|^2] \le \sigma^2 \qquad \text{for all } x. \tag{2.6}$$

Transport instead uses a condition on gradient differences.

Assumption A4 (Centered δ-similarity)**.** For some $\delta \ge 0$,

$$\mathbb{E}_\xi[\|r(x, \xi) - r(y, \xi)\|^2] \le \delta^2\|x - y\|^2 \qquad \text{for all } x, y. \tag{2.7}$$

Under uniform finite-sum sampling, this condition coincides with the Hessian-variance condition of Szlendak et al. [19] and the second-order similarity assumption of Khaled and Jin [10]. More recently, Shulgin et al. [17] generalized it through γ-similarity, which recovers this condition at $\gamma = 1$.

We use the name *centered δ-similarity* to emphasize that it measures the variation of gradient noise rather than its magnitude.

Here σ bounds the magnitude of individual gradient errors, whereas δ measures their variation under shared sampling. A small δ means that stochastic gradient differences closely match deterministic gradient differences; it does not require the gradients themselves, or their changes, to be small. Similarity is a condition on the stochastic oracle and applies to both finite-sum and population objectives.

For a finite sum $f = \frac{1}{n}\sum_{i=1}^{n} f_i$ with quadratic components $f_i(x) = \frac{1}{2}x^\top A_i x + b_i^\top x + c_i$ and symmetric A_i, the linear terms $b_i^\top x$ cancel from gradient differences, so identical Hessians give exact transport even when individual gradients differ greatly. Hessian variation controls δ, whereas L concerns the average Hessian; neither must be smaller than the other. Appendix C.2 gives the calculation.

Scalar least squares separates the two quantities. Let $f(x) = \frac{1}{2}\mathbb{E}[(ux - b)^2]$, with feature u satisfying $0 < \mathbb{E}[u^2]$ and $\mathbb{E}[u^4] < \infty$, and label $b = ux^\star + \varepsilon$. The label noise ε is independent of u, with $\mathbb{E}[\varepsilon] = 0$ and $\mathbb{E}[\varepsilon^2] = s^2 < \infty$. The stochastic gradient is $h(x; u, \varepsilon) = u^2(x - x^\star) - u\varepsilon$, so

$$\begin{aligned} h(x; u, \varepsilon) - h(y; u, \varepsilon) &= u^2(x - y), \\ \mathbb{E}[|h(x; u, \varepsilon) - \nabla f(x)|^2] &= \delta^2(x - x^\star)^2 + Ls^2, \end{aligned} \tag{2.8}$$

where $\delta^2 = \mathrm{Var}(u^2)$ and $L = \mathbb{E}[u^2]$. Label noise increases the variance of fresh gradients but cancels from differences; feature variability sets δ. Increasing s therefore affects TSGD's sampled initialization but not its later increments. When $\delta > 0$, the pointwise noise variance is unbounded in x, so similarity holds but global BV fails. Sections 3.1 and 3.3 illustrate the regimes of exact transport ($\delta = 0$) and noiseless labels ($s = 0$), respectively.

SGD can also be analyzed under the ABC condition, which generalizes BV [11]. Both constrain the stochastic gradient at each point separately; centered similarity instead constrains noise differences at two points under shared sampling. Appendix C.1 compares these conditions.

2.3 Two Fundamental Sources of TSGD Error

The introduction identified two sources of TSGD error: initialization and transport. We now make this decomposition precise. Define $e_k = g_k - \nabla f(x_k)$, $E_k = \mathbb{E}[\|e_k\|^2]$, and $d_{k+1} = D_{k+1} - [\nabla f(x_{k+1}) - \nabla f(x_k)]$. The transport update gives

$$\boxed{e_{k+1} = e_k + d_{k+1}.} \tag{2.9}$$

Let $\mathcal{F}_k$ denote the history before drawing D_{k+1}, including x_{k+1} and g_k. Conditionally unbiased increments give

$$\mathbb{E}[g_{k+1} \mid \mathcal{F}_k] = \nabla f(x_{k+1}) + e_k.$$

Unlike SGD, TSGD is therefore generally conditionally biased: the new increment is unbiased, but the estimator inherits its previous error. The increment error d_{k+1} has zero conditional mean and is orthogonal in expectation to previous errors. Consequently,

$$E_k = E_0 + \sum_{t=0}^{k-1} \mathbb{E}[\|d_{t+1}\|^2]. \tag{2.10}$$

Even if $\mathbb{E}[e_0] = 0$, so that $\mathbb{E}[e_k] = 0$, the mean-square error E_k cannot decrease. Initialization error persists, while transport errors accumulate. Similarity bounds each contribution by the squared

displacement, yielding

$$E_k \le E_0 + \delta^2 \sum_{t=0}^{k-1} \eta_t^2 \mathbb{E}[\|g_t\|^2]. \tag{2.11}$$

This explains both the opportunity and the difficulty: accurate changes preserve useful information, but even small errors can accumulate. Section 4 states the conditional assumptions directly on D_{k+1}, allowing transport beyond pointwise gradient oracles.

3 When Does TSGD Shine?

Through explicit examples, we now examine when TSGD benefits from accurate gradient changes and when SGD benefits from fresh estimates. We measure stationarity by the expected squared true-gradient norm, $G_k := \mathbb{E}[\|\nabla f(x_k)\|^2]$, and show how initialization quality and noise variation affect the comparison.

Alongside exact calculations, we use a vector least-squares model $f(x) = \frac{1}{2}\mathbb{E}[\|Ax - b\|^2]$ as a running experiment. Its population Hessian is held fixed while feature variability sets δ and label noise varies independently (Section 2.2).

3.1 Accurate Differences Despite Noisy Gradients

Consider $f(x) = x^2/2$ and an unbiased oracle $h(x, \xi) = x + \xi$, where $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[\xi^2] = \sigma^2$. The individual gradients can be arbitrarily noisy, yet

$$h(x, \xi) - h(y, \xi) = x - y = \nabla f(x) - \nabla f(y).$$

Thus $\delta = 0$, independently of σ. With exact initialization, TSGD follows gradient descent exactly. Indeed, $g_k = \nabla f(x_k)$ and an exact difference give

$$g_{k+1} = g_k + \nabla f(x_{k+1}) - \nabla f(x_k) = \nabla f(x_{k+1}).$$

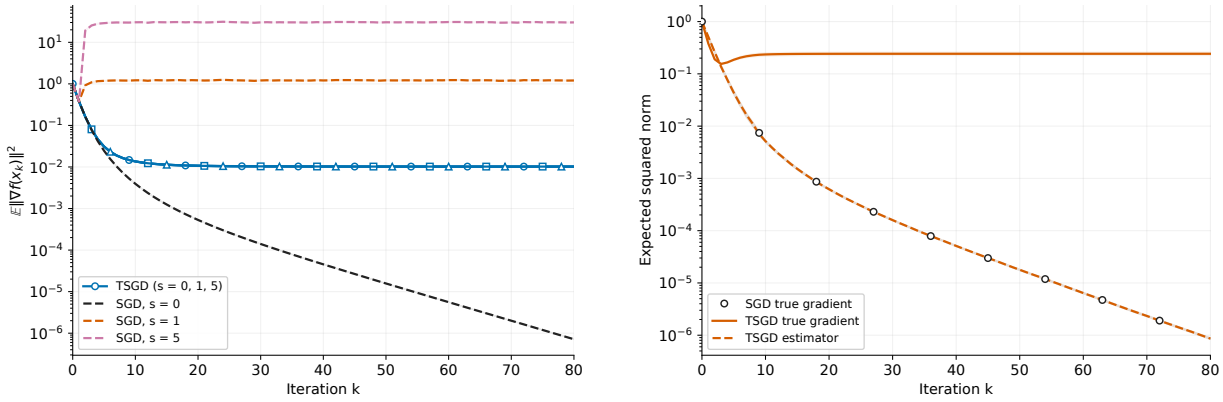
Induction from $g_0 = \nabla f(x_0)$ proves this for any objective. Appendix B.2 gives the resulting GD convergence guarantee and proves that the gradients tend to zero.

For the present quadratic, take $x_0 = 1$, $\eta = 1$, and any initial estimate $g_0 = 1 + e_0$ with $E_0 = \mathbb{E}[e_0^2] < \infty$, biased or not. Then $x_1 = -e_0$ and $g_1 = g_0 + (x_1 - x_0) = 0$, so TSGD stays at $x_k = -e_0$ for all $k \ge 1$. SGD instead uses a fresh gradient at every update, giving $x_{k+1} = -\xi_k$. Consequently,

$$\boxed{G_k^{\mathrm{TSGD}} = E_0, \qquad G_k^{\mathrm{SGD}} = \sigma^2, \qquad k \ge 1.}$$

Transport preserves the quality of its initial information, good or bad: TSGD has the smaller error when $E_0 < \sigma^2$ and the larger error when $E_0 > \sigma^2$. A single exact gradient evaluation at initialization, $g_0 = \nabla f(x_0)$, gives $E_0 = 0$: TSGD reaches the minimizer in one step and $G_k^{\mathrm{TSGD}} = 0$ for every $k \ge 1$, without any further exact gradient evaluations. In contrast, single-sample initialization $g_0 = h(x_0, \xi_0)$ gives the same error as SGD.

Figure 1(a) illustrates the same separation with varying features. Holding the initial estimate and feature draws fixed, TSGD's trajectories coincide across label-noise levels, while SGD's error floor increases with the label noise. TSGD nevertheless plateaus because $\delta > 0$ and transport errors accumulate at a constant stepsize; without label noise, SGD eventually overtakes it, as the fresh-gradient example in Section 3.3 also illustrates.



(a) Label noise cancels from transport.

(b) False convergence of the transported estimate.

Figure 1: Least-squares experiments with a common stepsize. (a) With exact initialization and matched feature draws, TSGD's trajectories coincide across label-noise levels, while SGD's error floor grows with the noise. (b) With noiseless labels and variable features, SGD's true gradient decays to zero, while TSGD's estimator vanishes but its true gradient does not. Curves show means over 4,000 trajectories with pointwise bands of ± 2 standard errors; the appendix gives the full setup.

3.2 Fixed Noise Magnitude, Variable Similarity

To determine whether noise magnitude alone explains the relative behavior of SGD and TSGD, we compare oracles with the same pointwise noise distribution but different noise variation. For $\omega \geq 0$, consider

$$f(x) = \frac{L}{2}x^2, \qquad h(x,\varphi) = Lx + \sqrt{2}\sigma\cos(\omega x + \varphi), \qquad \varphi \sim \mathrm{Unif}[0, 2\pi). \tag{3.1}$$

Each new sample is an independent phase; a transport difference uses the same phase at both endpoints. For the noise $r(x,\varphi) = h(x,\varphi) - Lx$,

$$\mathbb{E}[r(x,\varphi)^2] = \sigma^2, \qquad \mathbb{E}[|r(x,\varphi) - r(y,\varphi)|^2] = 4\sigma^2 \sin^2\left(\frac{\omega(x-y)}{2}\right). \tag{3.2}$$

The smallest global similarity constant is $\delta = \sigma\omega$ (Appendix D.4). Pointwise noise has the same distribution for every x and ω, so SGD has identical expected squared-gradient dynamics at every frequency. TSGD, however, has exact transport at $\omega = 0$, while its similarity constant grows as $\delta = \sigma\omega$. Its actual transport error depends on the displacement through the sine term.

Thus pointwise noise distributions alone do not determine transport accuracy: they leave unchanged SGD's fresh estimates while permitting arbitrarily different similarity constants.

3.3 When Fresh Gradients Are Preferable

Now let

$$f(x) = \frac{x^2}{2}, \qquad h(x,a) = ax, \qquad \mathbb{P}(a=0) = \mathbb{P}(a=2) = \frac{1}{2}, \qquad x_0 = 1. \tag{3.3}$$

Here the gradient-noise variance is x^2: fresh gradients become exact at the minimizer, although global BV fails. Similarity holds with $\delta = 1$. At the common stepsize $\eta = 1/2$, each SGD sample either leaves the iterate unchanged or sends it to zero, so $G_k^{\mathrm{SGD}} = 2^{-k}$. In contrast, exactly initialized

Initialization	Exact transport	Stochastic transport
Exact	Gradient descent	Accumulated transport error
Inexact	Persistent initial error	Both errors and their interaction

Table 1: Four regimes in the quadratic model, with a stable constant stepsize. The exact-transport identities also hold for general objectives; the explicit limiting errors and stability condition are established in Appendix B.

TSGD satisfies $g_{k+1} = (1 - a_{k+1}/2)g_k$. Its estimator becomes zero at the first sample $a_{k+1} = 2$, wherever the iterate happens to be. The exact calculation in Appendix D.2 gives

$$g_k \longrightarrow 0 \quad \text{almost surely}, \qquad G_k^{\mathrm{TSGD}} \longrightarrow \frac{1}{2}.$$

This is false convergence: the estimator tends to zero while the expected squared norm of the true gradient stays positive. Fresh information can therefore be preferable even when initialization is exact.

Figure 1(b) shows the same phenomenon on least squares with noiseless labels and highly variable features. SGD's true gradient decays in mean square, as does TSGD's estimator, but TSGD's expected squared true-gradient norm remains positive. A small transported estimate therefore does not certify a small true gradient.

3.4 What Transport Cannot Do

To separate the effects of initialization and transport, consider $f(x) = \lambda x^2/2$ and $h(x; a, b) = (\lambda + a)x + b$, with $\lambda > 0$, independent noises satisfying $\mathbb{E}[a] = \mathbb{E}[b] = 0$, $\mathbb{E}[a^2] = \delta^2$, and $\mathbb{E}[b^2] < \infty$. Shared sampling cancels b. Table 1 summarizes the four cases, whose exact dynamics are developed in Appendix B.

Exact transport preserves initialization error. The behavior in Section 3.1 extends beyond quadratics. When $\delta = 0$, the error identity gives $e_k = e_0$, hence $g_k = \nabla f(x_k) + e_0$ for every k. With $e_0 = 0$ this is GD; otherwise it is GD on $f(x) + \langle e_0, x\rangle$. The reason is that adding a linear term leaves all gradient differences unchanged, so differences alone cannot recover a missing constant gradient offset (Appendix B.2).

Stochastic transport can produce a nonzero error floor. For $0 < \eta < 2\lambda/(\lambda^2 + \delta^2)$ and $M_0 := \mathbb{E}[|g_0|^2]$, the exact quadratic dynamics give

$$\lim_{k\to\infty} \mathbb{E}[|g_k|^2] = 0, \qquad \lim_{k\to\infty} G_k = E_0 + \frac{\eta\delta^2 M_0}{2\lambda - \eta(\lambda^2 + \delta^2)}. \tag{3.4}$$

With exact initialization, $M_0 = \lambda^2 x_0^2$: if $\delta > 0$ and $x_0 \neq 0$, the estimate vanishes in mean square while the true gradient has a positive limiting mean-square norm. Appendix B.3 proves both limits.

The two errors interact. Initialization affects the residual directly through E_0 and indirectly through M_0, which controls the steps and the transport errors they generate. For centered initialization error, $M_0 = \lambda^2 x_0^2 + E_0$, so improving the initial estimate reduces both contributions. Appendix B.3 derives this interaction and the associated conservation law. Appendix D.5 varies initialization accuracy and similarity on least squares and compares the observed residuals with their exact limits.

4 Convergence Theory and Comparison with SGD

The examples distinguish noise magnitude from noise variation. We now make this distinction explicit in finite-horizon stationarity guarantees. Using G_k from the preceding section, write

$$\Delta_0 := f(x_0) - f_{\inf}, \qquad \overline{G}_K := \frac{1}{K}\sum_{k=0}^{K-1} G_k,$$

with a superscript identifying the algorithm when needed.

4.1 Main Convergence Guarantee for TSGD

Using the history $\mathcal{F}_k$ defined in Section 2.3, write $m_k := \nabla f(x_{k+1}) - \nabla f(x_k)$. We impose two conditions on the increments actually used along the trajectory.

Assumption A5 (Unbiased transport)**.** The increments satisfy

$$\mathbb{E}[D_{k+1} \mid \mathcal{F}_k] = m_k. \tag{4.1}$$

Assumption A6 (Similarity along the trajectory)**.** For some $\delta \geq 0$,

$$\mathbb{E}[\|D_{k+1} - m_k\|^2 \mid \mathcal{F}_k] \leq \delta^2 \|x_{k+1} - x_k\|^2. \tag{4.2}$$

Assumptions A2 and A4 imply these two conditions for shared-sample gradient differences. The conditional similarity requirement may be weaker than global similarity because it concerns only the generated increments. Stating them directly on the increments also allows transport without a pointwise gradient oracle.

Theorem 4.1 (Finite-horizon stationarity of TSGD)**.** *Suppose the objective is smooth and lower bounded (Assumption A1), and the transport increments are conditionally unbiased and satisfy centered δ-similarity (Assumptions A5–A6). Allow any initial estimate with $E_0 < \infty$, including biased ones. For an integer $K \geq 1$, choose a constant stepsize $\eta > 0$ satisfying*

$$L\eta + (K-1)\delta^2\eta^2 \leq 1. \tag{4.3}$$

Then

$$\overline{G}_K^{\mathrm{TSGD}} \leq \frac{2\Delta_0}{\eta K} + E_0. \tag{4.4}$$

Why the guarantee holds. Let $F_k := \mathbb{E}[f(x_k) - f_{\inf}]$ and $M_k := \mathbb{E}[\|g_k\|^2]$. Smoothness gives

$$F_{k+1} \leq F_k - \frac{\eta}{2}G_k - \frac{\eta}{2}(1 - L\eta)M_k + \frac{\eta}{2}E_k. \tag{4.5}$$

The error identity (2.10) and similarity imply $E_k \leq E_0 + \delta^2\eta^2 \sum_{t<k} M_t$. When the descent inequalities are summed, an error generated at step t enters at most $K-1-t$ later estimates. Condition (4.3) ensures that its total contribution is no larger than the corresponding negative M_t term. This leaves the initialization error and the initial objective gap in (4.4). Appendix A gives the full proof, retaining the terminal objective gap and all estimator-norm terms. Neither bounded variance nor a pointwise gradient oracle is required.

	SGD	TSGD
Noise assumption	BV: pointwise magnitude σ	Centered similarity: variation δ
Tuned bound on $\overline{G}_K$	$\frac{2L\Delta_0}{K} + \sqrt{\frac{2L\Delta_0\sigma^2}{K}}$	$\frac{2L\Delta_0}{K} + \frac{2\delta\Delta_0}{\sqrt{K}} + E_0$
Pointwise evaluations	K	$B_0 + 2(K-1)$

Table 2: Finite-horizon guarantees under each method's noise assumptions, with constant stepsizes optimized separately. Both require a smooth, lower-bounded objective. Costs count K iterate updates and an initialization using B_0 evaluations for TSGD.

Choosing the largest admissible stepsize gives

$$\boxed{\overline{G}_K^{\mathrm{TSGD}} \leq \frac{2L\Delta_0}{K} + \frac{2\delta\Delta_0}{\sqrt{K}} + E_0.} \tag{4.6}$$

The TSGD bound does not explicitly depend on σ, but this dependence may enter through E_0. For example, initializing g_0 by averaging B_0 independent unbiased stochastic gradients at x_0 gives $E_0 \leq \sigma^2/B_0$ under bounded variance. Figure 2 illustrates how a more accurate sampled initial estimate lowers TSGD's residual error. Thus, TSGD's sensitivity to gradient-noise magnitude depends on how its initial estimate is constructed.

The three terms represent deterministic optimization, transport noise, and initialization error. Small δ permits larger steps; $\delta = E_0 = 0$ recovers GD at $\eta = 1/L$. The coefficient of E_0 is sharp (Appendix B). For $\delta > 0$, the theorem requires a smaller step for a longer horizon; it does not assert asymptotic stationarity for an infinite fixed-step run.

Related single-run guarantees appear in SARAH-IN, iSARAH-IN, and PAGE-NS [14, 15, 18]. The formulation here expresses the argument directly through unbiased increments and their centered similarity.

4.2 Comparison with SGD

For SGD, unbiasedness and BV give the standard smooth nonconvex guarantee [6]. The fixed step $\eta = 1/L$ yields $\overline{G}_K^{\mathrm{SGD}} \leq 2L\Delta_0/K + \sigma^2$. For $\Delta_0 > 0$, optimizing the constant step for the prescribed horizon replaces this persistent variance term by a decaying one, as shown in Table 2. The exact stepsizes and derivations are in Appendix A.

The comparison reflects the different information the methods use: SGD's bound depends on noise magnitude, while TSGD's depends on noise variation and initial accuracy. Small δ can favor transport even when σ is large. Neither BV nor the more general ABC condition is comparable with global similarity (Appendix C.1), so these bounds apply under different assumptions and do not establish universal superiority.

4.3 Computational Considerations

Shared-sample transport uses two gradient evaluations per increment, but accurate differences may require fewer samples than accurate absolute gradients. Its cost must be considered together with that of initialization. Appendix E compares the guarantees at equal oracle budgets rather than only equal iterations.

5 TSGD as a Building Block of Modern Optimizers

5.1 Transport Is Already Everywhere

Table 3 shows how modern optimizers combine fresh estimation, averaging, and transport. Here H_{k+1} is a fresh unbiased estimate of $\nabla f(x_{k+1})$, and D_{k+1} is a shared-sample gradient difference. The weights $\alpha, \beta \in [0, 1]$ control fresh estimation, while γ scales MARS's transport correction.

Method	Estimator at iteration $k+1$	Role of transport
SARAH / SPIDER	H_{k+1} or $g_k + D_{k+1}$	Transport between periodic refreshes.
Geom-SARAH	H_{k+1} or $g_k + D_{k+1}$	Transport for a geometric number of steps.
PAGE	H_{k+1} or $g_k + D_{k+1}$	Refresh with probability p.
MARINA	$\nabla f(x_{k+1})$ or $g_k + D^{\mathcal{Q}}_{k+1}$	Communicate compressed gradient changes.
STORM / MVR	$(1-\alpha)(g_k + D_{k+1}) + \alpha H_{k+1}$	Average fresh and transported estimates.
MARS (unclipped)	$(1-\beta)(g_k + \gamma D_{k+1}) + \beta H_{k+1}$	Scale the transport correction within momentum.

Table 3: Transport in modern optimizers. Rows with two alternatives select one branch according to a restart or refresh rule; the last two rows form weighted combinations. $D^{\mathcal{Q}}_{k+1}$ denotes an unbiased compressed increment. The identities describe estimators; batch sizes, stepsizes, and iterate updates belong to each algorithm.

SARAH and SPIDER use transport between gradient recomputations [4, 13, 14]; Inexact SARAH allows batch initialization [15], and Geom-SARAH randomizes phase lengths [8]. PAGE refreshes probabilistically [12], while MARINA transports compressed differences [7]. MVR/STORM averages fresh and transported estimates [2]; unclipped MARS includes MSGD at $\gamma = 0$ and MVR at $\gamma = 1$ [20]. These identities concern the estimators before clipping or preconditioning; cached gradients need not satisfy the shared-sample transport assumptions.

SVRG [9] and SAGA [3] also exploit gradient differences, but do not transport an estimate along successive iterates. SVRG uses an anchor point and its full gradient; SAGA uses stored component gradients and their average. Their standard estimates are conditionally unbiased, whereas TSGD inherits tracking error. Gradient differences also appear in decentralized gradient tracking, where agents combine local gradient changes with neighbors' information to track the network's average gradient [16].

Practical relevance. SARAH and SARAH+ were evaluated on MNIST and CIFAR-10 classifiers [14], PAGE on LeNet, VGG, and ResNet [12], and MARS on GPT-2 language models [20]. These experiments use composite optimizers, not uninterrupted TSGD: even the paper analyzing SARAH-IN uses restarted SARAH/SARAH+ in its neural-network experiments. Some practical MARS variants also use different samples in their gradient corrections. These results do not establish pure TSGD's performance in isolation.

5.2 Why Combine Fundamental Estimators?

Modern optimizers combine fresh estimation, averaging, and transport through weighted averages or refreshes. Three considerations motivate these combinations.

Computational efficiency. Accurate gradient changes may require far fewer samples than accurate absolute gradients. Transport can then preserve an expensive initial estimate through many cheaper steps, despite using two evaluations per sample. More frequent refreshes limit error accumulation but reduce this saving.

Complementary strengths. Momentum averages past estimates to reduce noise, but can lag behind a rapidly changing gradient. Transport accounts for this change, while fresh gradients correct inherited and accumulated errors at the cost of introducing pointwise noise. MVR, STORM, and MARS combine these roles even in a fixed noise regime.

Changing regimes. Shorter steps tighten the similarity bound on new transport errors without removing old ones. Fresh gradients may also improve: their noise vanishes near the minimizer in Section 3.3, while transport retains a residual. The relative reliability of the two information sources can therefore change during a run. Choosing refresh frequencies or mixing weights requires balancing their accuracy and cost, generally without knowing their errors.

Improved complexity through refreshing. When BV and centered similarity hold together, SPIDER [4] and PAGE [12] exploit both assumptions by combining accurate gradient recomputations with cheaper transport. Refreshes limit how long transport errors accumulate, allowing larger steps without an expensive estimate at every iteration. Their prescribed batch sizes and refresh schedules give $O(\varepsilon^{-3})$ oracle complexity, compared with the $O(\varepsilon^{-4})$ guarantees for SGD and pure TSGD in Section 4. The exponent three matches the lower bound of Fradin et al. [5] for zero-respecting algorithms on the same class (Appendix E.2). This comparison motivates combining estimators; it is not an $\Omega(\varepsilon^{-4})$ lower bound for pure TSGD.

6 Conclusion

Transported SGD is a fundamental way to use stochastic-gradient information: it estimates changes and carries an existing estimate along the trajectory. With a sufficiently accurate initial estimate, small variation in gradient noise can make transport effective even when individual gradients are very noisy. Its performance depends on the initialization error it retains and the transport errors it accumulates. The recursion is already studied on its own in methods such as SARAH-IN, and also appears within more elaborate optimizers. Our presentation emphasizes its role as a fundamental estimator and makes its two error sources explicit. We also briefly discuss reasons for combining transport with fresh estimation and momentum averaging.

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A Detailed Convergence Proofs

A.1 Descent and Weighted Accumulation

Fix deterministic x_0, horizon $K \geq 1$, and constant step $\eta > 0$. Finite second moments follow inductively from smoothness, similarity, and $E_0 < \infty$. Write $q_k = \mathbb{E}[\|d_{k+1}\|^2]$. Conditional

unbiasedness gives $\mathbb{E}[\langle e_k, d_{k+1}\rangle] = 0$, so $E_{k+1} = E_k + q_k$. Smoothness and $2\langle \nabla f(x_k), g_k\rangle = \|\nabla f(x_k)\|^2 + \|g_k\|^2 - \|e_k\|^2$ give (4.5). Summing the error identity yields

$$\sum_{k=0}^{K-1} E_k = KE_0 + \sum_{t=0}^{K-2} (K-1-t)q_t, \tag{A.1}$$

where the sum is empty for $K = 1$, and similarity gives

$$q_t \le \delta^2 \mathbb{E}[\|x_{t+1} - x_t\|^2] = \delta^2\eta^2 M_t. \tag{A.2}$$

A.2 Proof of Theorem 4.1

Sum (4.5), substitute (A.1)–(A.2), and collect the coefficients of M_k. This gives

$$\frac{1}{K}\sum_{k=0}^{K-1} G_k + \frac{2F_K}{\eta K} + \frac{1}{K}\sum_{k=0}^{K-1} c_k M_k \le \frac{2\Delta_0}{\eta K} + E_0, \tag{A.3}$$

where $c_k = 1 - L\eta - \delta^2\eta^2(K-1-k)$. The stepsize condition makes every c_k nonnegative. Dropping these terms and $F_K \ge 0$ proves the theorem.

A.3 Exact Optimized TSGD Guarantee

The largest positive step satisfying $L\eta + (K-1)\delta^2\eta^2 \le 1$ is

$$\boxed{\eta^\star = \frac{2}{L + \sqrt{L^2 + 4(K-1)\delta^2}}.} \tag{A.4}$$

This is the positive quadratic root when $(K-1)\delta^2 > 0$ and equals $1/L$ otherwise. It optimizes the sufficient condition, not the actual stability limit. Substitution into (4.4) yields

$$\overline{G}_K^{\mathrm{TSGD}} \le \frac{\Delta_0}{K}\left(L + \sqrt{L^2 + 4(K-1)\delta^2}\right) + E_0. \tag{A.5}$$

Using

$$\sqrt{L^2 + 4(K-1)\delta^2} \le L + 2\delta\sqrt{K-1}, \qquad \frac{\sqrt{K-1}}{K} \le \frac{1}{\sqrt{K}},$$

gives (4.6).

A.4 SGD under Bounded Variance

Under Assumptions A1–A3, let F_k, G_k now refer to the SGD iterates. Conditional unbiasedness and BV imply

$$\mathbb{E}_\xi[\|h(x_k,\xi)\|^2 \mid x_k] \le \|\nabla f(x_k)\|^2 + \sigma^2.$$

Consequently,

$$F_{k+1} \le F_k - \eta\left(1 - \frac{L\eta}{2}\right) G_k + \frac{L\eta^2\sigma^2}{2}.$$

Summing and dropping $F_K \ge 0$ yields, for $0 < \eta < 2/L$,

$$\boxed{\overline{G}_K^{\mathrm{SGD}} \le \frac{2\Delta_0 + KL\eta^2\sigma^2}{\eta K(2 - L\eta)}.} \tag{A.6}$$

In particular, the fixed gradient-descent step $\eta = 1/L$ gives

$$\overline{G}_K^{\mathrm{SGD}} \le \frac{2L\Delta_0}{K} + \sigma^2. \tag{A.7}$$

For $\Delta_0 > 0$, differentiating the bound shows that its minimum satisfies $L\sigma^2\eta^2 = (2\Delta_0/K)(1-L\eta)$, giving

$$\boxed{\eta^*_{\mathrm{BV},K} = \frac{2}{L + \sqrt{L^2 + 2KL\sigma^2/\Delta_0}}.} \tag{A.8}$$

At this step the bound equals $2\Delta_0/(K\eta)$, hence

$$\boxed{\overline{G}_K^{\mathrm{SGD}} \le \frac{\Delta_0}{K}\left(L + \sqrt{L^2 + \frac{2KL\sigma^2}{\Delta_0}}\right) \le \frac{2L\Delta_0}{K} + \sqrt{\frac{2L\Delta_0\sigma^2}{K}}.} \tag{A.9}$$

The last inequality uses $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$. For $\sigma = 0$ the step is $1/L$; for $\sigma > 0$ it decreases with K. If $\Delta_0 = 0$, the initial point is already stationary and the optimized formula is not used.

A.5 Iteration Complexity

Fix an accuracy target $\varepsilon > 0$ and let R be uniform on $\{0, \dots, K-1\}$, independent of the run. Then $\mathbb{E}[\|\nabla f(x_R)\|^2] = \overline{G}_K$.

Corollary A.1 (Sufficient TSGD horizon)**.** *Under Theorem 4.1, suppose $E_0 \le \varepsilon^2/3$ and use $\eta = \eta^\star$. An integer horizon satisfying*

$$K \ge \max\left\{1, \frac{6L\Delta_0}{\varepsilon^2}, \frac{36\delta^2\Delta_0^2}{\varepsilon^4}\right\} \tag{A.10}$$

ensures $\mathbb{E}[\|\nabla f(x_R)\|^2] \le \varepsilon^2$.

Proof. Each of the first two terms in (4.6) is at most $\varepsilon^2/3$ under (A.10); the initialization term satisfies the same bound by assumption. □

For SGD with $\Delta_0 > 0$ and step (A.8), the analogous sufficient horizon is

$$K \ge \max\left\{1, \frac{4L\Delta_0}{\varepsilon^2}, \frac{8L\Delta_0\sigma^2}{\varepsilon^4}\right\}. \tag{A.11}$$

Each of the two terms in the simplified SGD bound is then at most $\varepsilon^2/2$. These are iteration upper bounds; initialization and per-iteration oracle costs are accounted for in Appendix E.

B Exact Quadratic Dynamics and Error Sources

B.1 Model and Recursions

Let x_0 be deterministic and

$$f(x) = \frac{\lambda x^2}{2}, \qquad h(x; a, b) = (\lambda + a)x + b, \qquad \lambda > 0,$$

where a, b are independent, centered, and $\mathbb{E}[a^2] = \delta^2$, $\mathbb{E}[b^2] = \sigma^2 < \infty$. Use independent fresh pairs at successive steps, independently of the square-integrable g_0. Shared sampling cancels b, giving $D_{k+1} = (\lambda + a_{k+1})(x_{k+1} - x_k)$ and

$$g_{k+1} = (1 - \eta\lambda - \eta a_{k+1})g_k, \qquad e_{k+1} = e_k - \eta a_{k+1} g_k. \tag{B.1}$$

Thus conditional similarity holds with equality, independently of σ. Write $M_k = \mathbb{E}[|g_k|^2]$, $E_k = \mathbb{E}[|e_k|^2]$, and $G_k = \mathbb{E}[|\lambda x_k|^2]$.

B.2 Exact Transport

For any objective, exact differences preserve $e_k = e_0$. TSGD is therefore GD on $\widetilde{f}(x) = f(x) + \langle e_0, x\rangle$, and is ordinary GD when $e_0 = 0$. At $\eta = 1/L$ and $e_0 = 0$, descent gives, for every K,

$$\frac{1}{2L} \sum_{k<K} \|\nabla f(x_k)\|^2 \le \Delta_0.$$

This proves the usual $2L\Delta_0/K$ averaged guarantee and, by boundedness of the nonnegative partial sums, $\sum_{k\ge 0} \|\nabla f(x_k)\|^2 < \infty$. Hence each gradient tends to zero, without convexity or convergence of the iterates. For fixed nonzero e_0, the same argument on $\widetilde{f}$ gives $\nabla f(x_k) \to -e_0$ if $\widetilde{f}$ is also bounded below. The finite-horizon bound $2L\Delta_0/K + E_0$ requires no such additional lower bound.

For the quadratic, put $\delta = 0$ and $r = 1 - \eta\lambda$. Equation (B.1) and $\lambda x_k = g_k - e_0$ give

$$g_k = r^k(\lambda x_0 + e_0), \qquad x_k = r^k x_0 - \frac{1 - r^k}{\lambda} e_0. \tag{B.2}$$

For centered e_0, $G_k = r^{2k}\lambda^2 x_0^2 + (1 - r^k)^2 E_0$. For any square-integrable e_0, biased or not, $|r| < 1$ gives $x_k \to -e_0/\lambda$ in mean square and $G_k \to E_0$.

Sharpness of the initialization coefficient. Choose $E_0 > 0$ and $\eta = 1/\lambda$. Then $G_k = E_0$ for every $k \ge 1$, so

$$\frac{1}{K} \sum_{k<K} G_k = \frac{\lambda^2 x_0^2}{K} + \left(1 - \frac{1}{K}\right) E_0 \longrightarrow E_0.$$

The bound $2\Delta_0/(\eta K) + E_0$ has the same limit. Thus the coefficient one on E_0 is sharp.

B.3 Stochastic Transport and the Interaction Formula

For arbitrary square-integrable initialization, set $\rho = (1 - \eta\lambda)^2 + \eta^2\delta^2$. Independence and centering in (B.1) give

$$M_{k+1} = \rho M_k, \qquad E_{k+1} = E_k + \eta^2\delta^2 M_k. \tag{B.3}$$

Consequently,

$$M_k = \rho^k M_0, \qquad E_k = E_0 + \eta^2\delta^2 M_0 \sum_{t<k} \rho^t.$$

Assume the mean-square stability condition

$$0 < \eta < \frac{2\lambda}{\lambda^2 + \delta^2}, \qquad \text{equivalently } \rho < 1. \tag{B.4}$$

With $A := \eta^2\delta^2/(1 - \rho)$, $A(1 - \rho) = \eta^2\delta^2$ and (B.3) imply the conservation law and its limit:

$$E_k + AM_k = E_0 + AM_0, \qquad E_\infty := \lim_k E_k = E_0 + AM_0. \tag{B.5}$$

To identify the limiting true-gradient error, use the orthogonality of the martingale increments. For $m > n$,

$$\mathbb{E}[|e_m - e_n|^2] = \eta^2\delta^2 \sum_{t=n}^{m-1} M_t \le \frac{\eta^2\delta^2 M_0}{1-\rho}\rho^n \to 0.$$

Thus (e_k) is Cauchy in the complete space L^2 and converges in mean square to e_∞. Since $g_k \to 0$ in mean square, $\lambda x_k = g_k - e_k \to -e_\infty$ in mean square, proving

$$\boxed{\lim_k G_k = E_\infty = E_0 + \frac{\eta\delta^2}{2\lambda - \eta(\lambda^2+\delta^2)} M_0.} \tag{B.6}$$

In particular, exact initialization gives $E_0 = 0$ and $M_0 = \lambda^2 x_0^2$. If $\delta > 0$ and $x_0 \neq 0$, the estimator vanishes in mean square while the true gradient has a positive limiting mean-square norm.

Initialization contributes directly through E_0 and indirectly through $M_0 = \lambda^2 x_0^2 + 2\lambda x_0 \mathbb{E}[e_0] + E_0$. Bias may increase or decrease the latter. For centered e_0,

$$E_\infty = \underbrace{E_0}_{\text{initialization}} + \underbrace{A\lambda^2 x_0^2}_{\text{transport}} + \underbrace{AE_0}_{\text{interaction}}. \tag{B.7}$$

This yields the four cases in Table 1. The stochastic-transport formulas use the quadratic model; the exact-transport identities hold for general objectives.

C Noise Assumptions

C.1 Bounded Variance, ABC, and Centered Similarity

Write $h(x,\xi) = \nabla f(x) + r(x,\xi)$ with $\mathbb{E}_\xi[r(x,\xi)] = 0$.

ABC generalizes BV. BV requires the gradient-noise variance to be uniformly bounded over the domain. The more general ABC condition, introduced as expected smoothness by Khaled and Richtárik [11], assumes that some $A, B, C \ge 0$ satisfy

$$\mathbb{E}_\xi[\|h(x,\xi)\|^2] \le 2A(f(x) - f_{\inf}) + B\|\nabla f(x)\|^2 + C \qquad \text{for all } x. \tag{C.1}$$

By unbiasedness, BV gives $(A, B, C) = (0, 1, \sigma^2)$. ABC allows the second moment to grow with the objective gap and gradient norm. If $C = 0$, and both $f(x_k) - f_{\inf} \to 0$ and $\|\nabla f(x_k)\|^2 \to 0$, then $\mathbb{E}_\xi[\|h(x_k,\xi)\|^2] \to 0$: the second-moment penalty in SGD's descent bound can vanish.

For example, let $f(x) = x^2/2$ and $h(x,a) = ax$, where a equals 0 or 2 with equal probability. Its residual is $r(x,a) = h(x,a) - \nabla f(x) = (a-1)x$, so

$$\mathbb{E}[h(x,a)] = x, \qquad \mathbb{E}[|h(x,a)|^2] = 2x^2, \qquad \mathbb{E}[|r(x,a)|^2] = x^2.$$

Thus ABC holds with $(A, B, C) = (0, 2, 0)$, while global BV fails because the noise variance x^2 is unbounded as $|x| \to \infty$, even though fresh gradients are exact at the minimizer.

One-point and two-point conditions. BV and ABC constrain the stochastic gradient at each point separately. Centered similarity constrains noise differences under shared sampling. Section 3.2 shows that identical pointwise distributions can have different similarity constants. The following global counterexamples establish more: neither BV nor ABC implies similarity, and similarity implies neither.

Neither BV nor ABC implies global similarity. Take $f(x) = x^2/2$ and $h(x, \xi) = x + \xi \sin(x^2)$, where $\mathbb{P}(\xi = -1) = \mathbb{P}(\xi = 1) = 1/2$. The oracle is unbiased and

$$\mathbb{E}_\xi[|h(x, \xi) - \nabla f(x)|^2] = \sin^2(x^2) \le 1,$$

so BV holds with $\sigma^2 = 1$, and ABC holds with $(A, B, C) = (0, 1, 1)$. Under shared sampling,

$$\mathbb{E}_\xi[|r(x, \xi) - r(y, \xi)|^2] = \big(\sin(x^2) - \sin(y^2)\big)^2.$$

A finite similarity constant would imply

$$\begin{aligned} |\sin(x^2) - \sin(y^2)| &\le \delta |x - y| && \text{for every } x, y, \\ |2x \cos(x^2)| = \lim_{y \to x} \frac{|\sin(y^2) - \sin(x^2)|}{|y - x|} &\le \delta && \text{for every } x. \end{aligned}$$

However, at $x_n = \sqrt{2\pi n}$, $n \ge 1$, this would require

$$|2x_n \cos(x_n^2)| = 2\sqrt{2\pi n} \le \delta \qquad \text{for every } n.$$

For any finite δ, choosing an integer $n > \delta^2/(8\pi)$ violates this inequality. Hence no finite global similarity constant exists.

Global similarity implies neither BV nor ABC. Take $f(x) = \log\cosh x$ and $h(x, \xi) = \tanh x + \xi x$, with $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[\xi^2] = \delta^2 > 0$. The objective is nonnegative and 1-smooth, since $f''(x) = 1/\cosh^2 x \in (0, 1]$. The unbiased oracle has residual $r(x, \xi) = \xi x$, and hence

$$\mathbb{E}_\xi[|r(x, \xi) - r(y, \xi)|^2] = \delta^2 (x - y)^2, \qquad \mathbb{E}_\xi[|r(x, \xi)|^2] = \delta^2 x^2.$$

Similarity holds with equality, but no finite global BV constant exists. Since $f_{\inf} = 0$, ABC would require

$$\tanh^2 x + \delta^2 x^2 \le 2A \log\cosh x + B \tanh^2 x + C \le 2A|x| + B + C \qquad \text{for all } x.$$

For $x \neq 0$, dividing by x^2 and dropping the nonnegative term $\tanh^2 x / x^2$ gives

$$\delta^2 \le \frac{2A}{|x|} + \frac{B + C}{x^2}.$$

For any finite nonnegative A, B, C, the right-hand side tends to zero as $|x| \to \infty$, whereas the left-hand side is the fixed positive constant δ^2. The inequality therefore fails for sufficiently large $|x|$, so no finite ABC constants exist.

For this oracle, Theorem 4.1 still applies: the coupled increment has conditional error variance $\delta^2 (x_{k+1} - x_k)^2$. Thus exact initialization gives the $O(K^{-1/2})$ TSGD guarantee despite failure of both BV and ABC. This does not by itself establish that SGD fails.

C.2 Quadratic Finite Sums and the Smallest Similarity Constant

For $f_i(x) = \frac{1}{2} x^\top A_i x + b_i^\top x + c_i$ with symmetric A_i, let $f = \frac{1}{n} \sum_{i=1}^n f_i$, $\bar{A} = \frac{1}{n} \sum_{i=1}^n A_i$, and sample i uniformly. The centered gradient difference is

$$[\nabla f_i(x) - \nabla f_i(y)] - [\nabla f(x) - \nabla f(y)] = (A_i - \bar{A})(x - y). \tag{C.2}$$

For $z = x - y$,

$$\mathbb{E}_i[\|(A_i - \bar{A})z\|^2] = z^\top \left(\frac{1}{n} \sum_{i=1}^{n} (A_i - \bar{A})^2 \right) z.$$

Taking the supremum over $\|z\| = 1$ gives the smallest admissible global constant,

$$\delta_{\min}^2 = \lambda_{\max} \left(\frac{1}{n} \sum_{i=1}^{n} (A_i - \bar{A})^2 \right). \tag{C.3}$$

The linear terms cancel irrespective of their magnitude, while the smoothness constant of the average quadratic is $L = \|\bar{A}\|_{\mathrm{op}}$. There is no fixed ordering between δ and L. Identical A_i give $\delta_{\min} = 0$ even with large L and arbitrarily different b_i. Conversely, for $n \geq 3$ scalar components, take $f_1(x) = nLx^2/2$ and $f_i(x) = 0$ for $i \geq 2$. Then $f(x) = Lx^2/2$ but

$$\delta_{\min}^2 = \frac{1}{n}((nL - L)^2 + (n-1)L^2) = (n-1)L^2 > L^2.$$

Thus similarity measures the discrepancy between stochastic and deterministic gradient changes, rather than the smoothness of their average.

D Detailed Illustrative Examples

D.1 Additive Noise: Exact Trajectories and Horizon Tuning

For explicit constants, specialize the additive example to

$$f(x) = \frac{x^2}{2}, \qquad h(x, \xi) = x + \xi, \qquad \mathbb{P}(\xi = 1) = \mathbb{P}(\xi = -1) = \frac{1}{2}, \qquad x_0 = 1. \tag{D.1}$$

Here $L = 1$, $\Delta_0 = 1/2$, $\sigma^2 = 1$, and $\delta = 0$. At $\eta = 1$, SGD satisfies $x_{k+1} = -\xi_k$, hence

$$G_k^{\mathrm{SGD}} = 1 \quad (k \geq 0), \qquad \boxed{\overline{G}_K^{\mathrm{SGD}} = 1.} \tag{D.2}$$

Thus the fixed-step noise floor is present in the actual trajectory.

For the horizon-dependent BV stepsize (A.8), write $\eta_K = 2/(1 + \sqrt{1 + 4K})$ and $r_K = 1 - \eta_K$. Independence and centeredness of the fresh sample give

$$G_{k+1}^{\mathrm{SGD}} = \mathbb{E}[(r_K x_k - \eta_K \xi_k)^2] = r_K^2 G_k^{\mathrm{SGD}} + \eta_K^2.$$

Solving from $G_0 = 1$ and averaging the geometric terms yields

$$G_k^{\mathrm{SGD}} = r_K^{2k} + \frac{\eta_K}{2 - \eta_K}(1 - r_K^{2k}),$$
$$\overline{G}_K^{\mathrm{SGD}} = \frac{\eta_K}{2 - \eta_K} + \left(1 - \frac{\eta_K}{2 - \eta_K}\right) \frac{1 - r_K^{2K}}{K(1 - r_K^2)}.$$

Since $\eta_K \sim K^{-1/2}$, $r_K^{2K} \leq e^{-2K\eta_K} \to 0$ and $1 - r_K^2 = \eta_K(2 - \eta_K) \sim 2K^{-1/2}$. Each term in the average is therefore asymptotic to $1/(2\sqrt{K})$:

$$\boxed{\overline{G}_K^{\mathrm{SGD}} = \frac{1 + o(1)}{\sqrt{K}}.} \tag{D.3}$$

The optimized BV bound is $(1+\sqrt{1+4K})/(2K)$, so both its rate and leading constant are attained by this prescribed run.

For TSGD, use $\eta = 1$ and any square-integrable $g_0 = 1 + e_0$. Exact differences give $D_{k+1} = x_{k+1} - x_k = -g_k$; therefore $g_1 = 0$ and $x_k = -e_0$ for all $k \geq 1$. Hence

$$\boxed{\overline{G}_K^{\mathrm{TSGD}} = \frac{1}{K} + \left(1 - \frac{1}{K}\right) E_0.} \tag{D.4}$$

With exact initialization, TSGD reaches the minimizer after one step. For fixed $E_0 > 0$, horizon-tuned SGD eventually has the smaller error. The comparison depends on initialization as well as transport.

D.2 Multiplicative Noise: Fresh Gradients and False Convergence

For (3.3), $\mathbb{E}[a] = 1$ and $\mathbb{E}[(a-1)^2] = 1$. The oracle is unbiased and its pointwise noise variance is x^2, so global BV does not hold. Coupled centered differences nevertheless satisfy

$$\mathbb{E}[|(a-1)(x-y)|^2] = |x-y|^2,$$

giving $\delta = 1$. These comparisons use exact dynamics rather than the BV theorem.

At $\eta = 1/2$, SGD obeys $x_{k+1} = (1 - a_k/2)x_k$; it remains at one until the first sample $a = 2$, then stays at zero. Thus

$$\boxed{G_k^{\mathrm{SGD}} = 2^{-k}, \qquad \overline{G}_K^{\mathrm{SGD}} = \frac{2(1-2^{-K})}{K}.} \tag{D.5}$$

For TSGD at the same step, let $g_0 = \nabla f(x_0) = 1$ and $N := \min\{n \geq 1 : a_n = 2\}$. The estimator recursion is $g_{k+1} = (1 - a_{k+1}/2)g_k$, so

$$\mathbb{P}(N = n) = 2^{-n}, \qquad g_k = \mathbf{1}_{\{k<N\}}, \qquad x_k = 1 - \frac{1}{2}\min\{k, N\}.$$

Since $N < \infty$ almost surely and $\mathbb{E}[N^2] = 6$, dominated convergence gives $x_k \to 1 - N/2$ in mean square. Using $\mathbb{E}[N] = 2$,

$$\boxed{\lim_{k\to\infty} G_k^{\mathrm{TSGD}} = \mathbb{E}[(1-N/2)^2] = \frac{1}{2}.} \tag{D.6}$$

The estimator becomes zero almost surely in finite time, but the limiting true gradient is nonzero with probability $1 - \mathbb{P}(N = 2) = 3/4$. Cesàro averaging also gives $\overline{G}_K^{\mathrm{TSGD}} \to 1/2$.

D.3 Sharpness for Horizon-Tuned TSGD

For the same oracle with exact initialization, (A.4) gives $\eta_K^\star = 2/(1+\sqrt{4K-3})$, since $L = \delta = 1$ and $\Delta_0 = 1/2$. The optimized guarantee becomes

$$\overline{G}_K^{\mathrm{TSGD}} \leq \frac{1+\sqrt{4K-3}}{2K}. \tag{D.7}$$

Lemma D.1 (Asymptotic sharpness for tuned TSGD)**.** *For (3.3), with exact initialization and the constant step $\eta_K^\star$ for the K-iteration run,*

$$\lim_{K\to\infty} \sqrt{K}\, \overline{G}_K^{\mathrm{TSGD}} = 1.$$

Proof. Fix $K \geq 2$ and write $\eta = \eta_K^\star$. The updates imply

$$g_{k+1} = (1 - \eta a_{k+1}) g_k, \qquad e_{k+1} = e_k + \eta(1 - a_{k+1}) g_k.$$

Set $C_k := \mathbb{E}[g_k e_k]$ and $q := \mathbb{E}[(1 - \eta a)^2] = 1 - 2\eta + 2\eta^2$. Independence yields

$$M_{k+1} = qM_k, \qquad E_{k+1} = E_k + \eta^2 M_k, \qquad C_{k+1} = (1 - \eta) C_k + \eta^2 M_k.$$

Starting from $M_0 = 1$ and $E_0 = C_0 = 0$, these give

$$M_k = q^k, \qquad E_k = \frac{\eta}{2(1-\eta)}(1 - q^k), \qquad C_k = \eta^2 \sum_{t=0}^{k-1} (1-\eta)^{k-1-t} q^t.$$

Since $0 < \eta < 1$ and $q < 1$, all summands are nonnegative and

$$\sum_{k<K} C_k \leq \eta^2 \left(\sum_{t \geq 0} q^t \right) \left(\sum_{j \geq 0} (1-\eta)^j \right) = \frac{\eta^2}{(1-q)\eta} = \frac{1}{2(1-\eta)}.$$

Thus the average cross moment is $O(K^{-1})$ as $\eta_K^\star \to 0$. For the other moments,

$$\frac{1}{K} \sum_{k<K} M_k = \frac{1 - q^K}{K(1-q)}, \qquad \frac{1}{K} \sum_{k<K} E_k = \frac{\eta}{2(1-\eta)} \left(1 - \frac{1}{K} \sum_{k<K} M_k \right).$$

Now $\eta_K^\star \sim K^{-1/2}$, $1 - q = 2\eta_K^\star(1 - \eta_K^\star) \sim 2K^{-1/2}$, and $q^K \leq e^{-K(1-q)} \to 0$. Therefore

$$\frac{1}{K} \sum_{k<K} M_k \sim \frac{1}{2\sqrt{K}}, \qquad \frac{1}{K} \sum_{k<K} E_k \sim \frac{1}{2\sqrt{K}}.$$

Finally, $x_k = g_k - e_k$ implies $G_k = M_k + E_k - 2C_k$. The averaged cross term is lower order, which proves the claim. □

The theorem's bound has the correct leading rate and constant for this prescribed TSGD run. Fixed-step SGD in (D.5) has average error $\Theta(K^{-1})$, whereas this horizon-tuned TSGD run has $\Theta(K^{-1/2})$ error. This is not a lower bound over all TSGD stepsize schedules.

D.4 Fixed Variance and Variable Similarity

Lemma D.2 (Fixed variance and variable similarity)**.** *For (3.1), the noise is centered, satisfies (3.2), and has smallest global similarity constant $\delta = \sigma\omega$ for $\omega \geq 0$.*

Proof. A uniform phase makes the cosine have mean zero and second moment $1/2$, so $\mathbb{E}[r(x,\varphi)^2] = 2\sigma^2(1/2) = \sigma^2$. For the coupled difference,

$$r(x,\varphi) - r(y,\varphi) = -2\sqrt{2}\sigma \sin\left(\frac{\omega(x+y)}{2} + \varphi \right) \sin\left(\frac{\omega(x-y)}{2} \right).$$

Squaring and using $\mathbb{E}_\varphi[\sin^2(c + \varphi)] = 1/2$ for every fixed c gives

$$\mathbb{E}[|r(x,\varphi) - r(y,\varphi)|^2] = 8\sigma^2 \cdot \frac{1}{2} \sin^2\left(\frac{\omega(x-y)}{2} \right) = 4\sigma^2 \sin^2\left(\frac{\omega(x-y)}{2} \right).$$

This verifies (3.2). The inequality $|\sin t| \leq |t|$ proves similarity with $\delta = \sigma\omega$; minimality follows from

$$\lim_{y \to x} \frac{4\sigma^2 \sin^2(\omega(x-y)/2)}{(x-y)^2} = \sigma^2\omega^2.$$

□

SGD's fresh noise has conditional variance σ^2 at every iterate, giving the frequency-independent exact recursion

$$G_{k+1}^{\mathrm{SGD}} = (1 - L\eta)^2 G_k^{\mathrm{SGD}} + L^2\eta^2\sigma^2. \tag{D.8}$$

Indeed, even its conditional noise distribution is independent of x_k and ω, since adding ωx_k to a uniform phase preserves its distribution modulo 2π. For TSGD,

$$\begin{aligned} E_{k+1} - E_k &= 4\sigma^2\mathbb{E}\left[\sin^2\left(\frac{\omega(x_{k+1} - x_k)}{2}\right)\right] \\ &\le \sigma^2\omega^2\eta^2 M_k. \end{aligned} \tag{D.9}$$

For fixed $\sigma > 0$, varying ω realizes any smallest global similarity constant without changing the pointwise noise distribution. The sine factor also shows that increasing ω need not increase every realized increment's error monotonically.

D.5 Least-Squares Experiments

We use one least-squares model to illustrate initialization quality, label-noise cancellation, false convergence, and the effect of similarity. The figures compare iterations at a common stepsize, not computational cost or optimally tuned performance.

Common setup. Let $f(x) = \frac{1}{2}\mathbb{E}[\|Ax - b\|^2]$, with

$$H = \mathrm{diag}(0.1, 0.2, \ldots, 1), \qquad A = \sqrt{v}\, H^{1/2}, \qquad b = s\varepsilon,$$

where $v = 1 + \kappa\zeta$, $\mathbb{P}(\zeta = 1) = \mathbb{P}(\zeta = -1) = 1/2$, and $\varepsilon \sim \mathcal{N}(0, I_{10})$ is independent of ζ. The minimizer is $x^\star = 0$, and

$$\nabla f(x) = Hx, \qquad h(x; v, \varepsilon) = vHx - s\sqrt{v}\, H^{1/2}\varepsilon, \qquad h(x; v, \varepsilon) - h(y; v, \varepsilon) = vH(x - y).$$

Thus label noise, controlled by s, cancels from shared-sample differences, while feature variability determines $\delta = \kappa$. We use $x_0 = \mathbf{1}/\|H\mathbf{1}\|$, $\eta = 0.5$, and 4,000 independent trajectories. The figures show the first 80 of 400 updates, with pointwise bands of ± 2 Monte Carlo standard errors. Initial estimates are independent of subsequent samples.

Initialization quality. In Figure 2, $s = 1$ and $\kappa = 0.05$: gradients are noisy but their differences are accurate. TSGD initializes by averaging $B_0 \in \{1, 128, 512\}$ actual stochastic gradients; SGD uses one fresh sample at every update. Single-sample initialization leaves TSGD with a larger residual than SGD, whereas the larger initial batches give smaller residuals. Accurate transport preserves useful initial information but does not correct a poor initial estimate. The extra cost of larger initial batches is not reflected in the iteration axis.

False convergence. Figure 1(b) uses noiseless labels ($s = 0$), $\kappa = 0.9$, and $B_0 = 128$ sampled gradients to initialize TSGD. SGD's true gradient and TSGD's estimator both tend to zero in mean square, but TSGD's true-gradient error remains positive. The plot therefore distinguishes a small estimator from actual stationarity.

Label-noise cancellation. In Figure 1(a), both methods instead start from the exact gradient. We fix $\kappa = 0.2$ and use matched feature draws across $s \in \{0, 1, 5\}$. TSGD's trajectories coincide because label noise cancels from every difference; its remaining error comes from feature variability. SGD's residual grows with label noise and vanishes at $s = 0$.

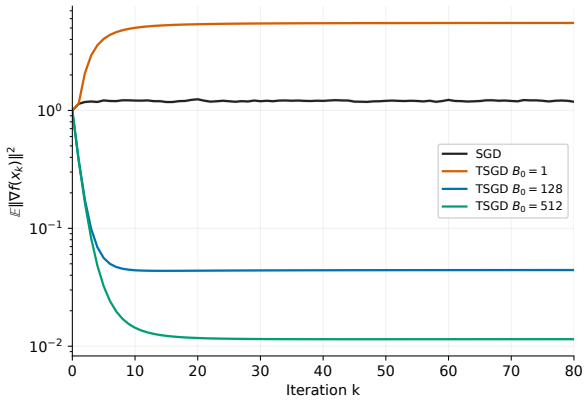


Figure 2: Sampled initialization: increasing B_0 lowers the residual when differences are accurate.

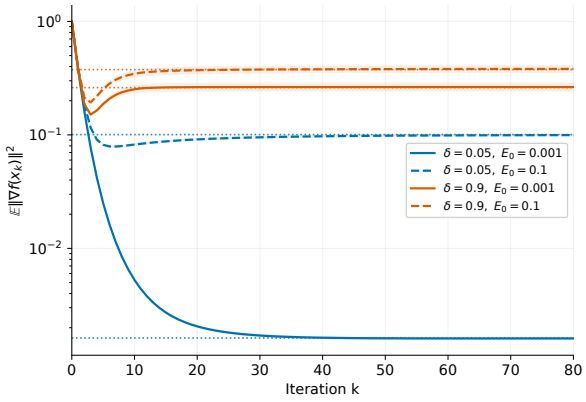


Figure 3: Controlled initialization: varying E_0 and δ. Dotted lines give the exact limits in (D.10).

Initialization and similarity. Figure 3 prescribes $g_0 = Hx_0 + e_0$, with $e_0 \sim \mathcal{N}(0, (E_0/10)I_{10})$, independently of subsequent samples. We vary $\delta = \kappa \in \{0.05, 0.9\}$ and $E_0 \in \{10^{-3}, 10^{-1}\}$. Applying (B.6) coordinatewise gives

$$\lim_k G_k = E_0 + \sum_{j=1}^{10} \frac{\eta\kappa^2\lambda_j}{2 - \eta(1+\kappa^2)\lambda_j} \left(\lambda_j^2 x_{0,j}^2 + \frac{E_0}{10}\right). \tag{D.10}$$

The denominators are positive for our parameters. The plateaus agree with these limits: better initialization reduces both inherited and transport error, while smaller δ reduces the transport contribution. This experiment and the label-noise experiment use controlled estimates, unlike the sampled initialization in the other two; they do not model how the initial accuracy is obtained.

E Oracle Complexity

E.1 Initialization and Evaluation Costs

Averaging B_0 independent unbiased gradients at x_0, independently of subsequent samples, gives $E_0 \le \sigma_0^2/B_0$, where σ_0^2 bounds the variance at x_0. Thus $B_0 \ge \max\{1, \lceil 3\sigma_0^2/\varepsilon^2 \rceil\}$ suffices for Corollary A.1. For K updates, SGD uses K evaluations and shared-sample TSGD uses $B_0 + 2(K-1)$; the last transport correction is unused. Combining these costs with the iteration bounds, under BV,

$$\begin{aligned} N_{\mathrm{SGD}}(\varepsilon) &= O\left(1 + \frac{L\Delta_0}{\varepsilon^2} + \frac{L\Delta_0\sigma^2}{\varepsilon^4}\right), \\ N_{\mathrm{TSGD}}(\varepsilon) &= O\left(1 + \frac{\sigma^2 + L\Delta_0}{\varepsilon^2} + \frac{\delta^2\Delta_0^2}{\varepsilon^4}\right). \end{aligned} \tag{E.1}$$

These are sufficient budgets, not lower bounds. At a common budget $N \ge \max\{1, B_0\}$, the corresponding horizons are $K_{\mathrm{SGD}} = N$ and $K_{\mathrm{TSGD}} = 1 + \lfloor (N - B_0)/2 \rfloor$. Initialization accuracy and cost must therefore both enter the comparison.

E.2 Why Refreshing Can Improve Complexity

When BV and centered similarity hold together, unbiasedness gives

$$\mathbb{E}[\|h(x,\xi) - h(y,\xi)\|^2] = \|\nabla f(x) - \nabla f(y)\|^2 + \mathbb{E}[\|r(x,\xi) - r(y,\xi)\|^2] \le (L^2 + \delta^2)\|x - y\|^2.$$

The mean-squared-smoothness guarantees of SPIDER and PAGE therefore give $O(\varepsilon^{-3})$ complexity on this same class [4, 12], with shared-sample access. The target is $\mathbb{E}[\|\nabla f(\widehat{x})\|] \le \varepsilon$; our squared-norm target implies it by Jensen's inequality.

Corollary E.1 (Fradin et al., Theorem 3.2, specialized)**.** *Fix positive bounds $\Delta, L, \sigma, \delta$ on the initial gap, smoothness, BV, and centered similarity. For sufficiently small $\varepsilon > 0$, zero-respecting algorithms require $\Omega(\Delta\delta\sigma/\varepsilon^3)$ evaluations in the worst case to achieve either target above. The oracle allows any fixed finite number of simultaneous shared-sample queries per round, with independent samples across rounds.*

Proof. Set $p = q = 2$, $L_1 = L$, and $\sigma_1 = \sigma$ in Fradin et al. [5, Theorem 3.2]. Their moment and similarity conditions become BV and centered similarity, and the bound is

$$\Omega\left(\min\left\{\frac{L\Delta}{\varepsilon^2} + \frac{L\Delta\sigma^2}{\varepsilon^4}, \frac{\sigma^2 + (L+\delta)\Delta}{\varepsilon^2} + \frac{\delta\Delta\sigma}{\varepsilon^3}\right\}\right).$$

For fixed positive parameters and sufficiently small ε, the second branch is smaller and its cubic term dominates. The theorem's restriction $\varepsilon \le c_1\sqrt{L\Delta}$ is then satisfied. Each query round requires at least one evaluation; Jensen's inequality extends the lower bound to the squared-norm target. □

Zero-respecting algorithms introduce only coordinates revealed by past oracle responses when started at the origin; this is not a lower bound for all randomized algorithms. It establishes optimality of the exponent three for this class, not an ε^{-4} lower bound for pure TSGD.